HCD^2: A Generative Dense Classification Framework for Hyperspectral Imagery With Conditional Diffusion

Hongqiao Wang, *Member, IEEE*, Haiyan Sun, Xuanchen Yan, Yanping Cai, and Zhaoxiang Zhang, *Member, IEEE*

***Abstract*—Hyperspectral image (HSI) classification aims to recover accurate land-cover semantic distributions from high-dimensional spectral–spatial observations remaining a critical pillar in intelligent Earth observation. Existing models, such as the convolutional neural networks (CNNs), the Transformers, and the Mamba are predominantly based on discriminative learning paradigms, which often struggle to effectively model the underlying spectral–spatial distributions of ground objects under complex noise, spectral mixing, and intraclass spectral variability. To address the limitations, a hyperspectral conditional diffusion for dense (HCD^2) classification framework is proposed in this article, which reformulates HSI classification as a conditional label-map denoising process guided by hyperspectral features, enabling generative dense semantic inference. Specifically, in the encoding stage, an adaptive spectral decoupling (ASD) module is designed to enhance class-discriminative spectral representations. Furthermore, a spatial–spectral joint topological encoder (SSJTE) is constructed to jointly model long-range spatial dependencies and spectral topological structures, thereby learning robust spectral–spatial representations. In the decoding stage, a conditional diffusion category decoder (CDCD) is proposed to achieve semantic category restoration and progressive refinement through conditional diffusion and iterative denoising. Experiments on several public datasets demonstrate that the proposed HCD^2 consistently outperforms existing discriminative and generative methods, while exhibiting robustness in complex scenarios.**

***Index Terms*—Conditional diffusion model, dense prediction, hyperspectral image (HSI) classification, spectral–spatial joint encoding.**

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I. Introduction

HYPERSPECTRAL remote sensing sensors can acquire spectral radiation information from hundreds of adjacent narrow spectral bands for ground objects. Each single pixel of the obtained hyperspectral image (HSI) matches an exclusive spectral response curve, which can effectively reflect the physical and chemical properties of different ground materials. Benefiting from such high-resolution spectral information, hyperspectral remote sensing has witnessed broad application prospects in various practical engineering tasks. Typical application fields include land cover classification, precision agricultural monitoring, ecological environment assessment, engineering geological survey, and military target reconnaissance, which are reported in existing research works [1], [2], [3], [4], [5], [6]. As one of the most core downstream tasks of hyperspectral data analysis, classification of HSI aims to assign each image pixel to its matched land cover category automatically. However, three inherent defects of original hyperspectral data severely hinder the classification performance in actual complex scenes: excessive spectral dimension, strong correlation between adjacent spectral bands, and the scarcity of manually annotated training samples [7].

At the initial study phase of HSI classification, researchers mainly adopted conventional machine learning algorithms to complete pixel-level classification tasks, including support vector machine, random forest, and sparse representation-based classification methods [8], [9]. These traditional algorithms can achieve stable classification results with limited labeled samples, and they also have advantages in computational efficiency. However, such methods rely heavily on manual feature engineering to extract effective spectral features. Customized shallow features cannot completely capture intricate nonlinear joint spatial–spectral correlations embedded in hyperspectral data, which becomes the main performance bottleneck of conventional classification models.

The booming advancement of deep learning techniques has greatly accelerated research progress in the field of HSI classification, and convolutional neural networks (CNNs) have become the mainstream basic model by virtue of powerful spatial–spectral joint feature extraction capability [10]. Subsequent studies further improve feature expression ability via

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combining 2-D and 3-D convolution modules and designing multiscale feature fusion mechanisms [11]. Even so, standard convolutional operations merely focus on limited local receptive areas, which fails to capture long-range spatial and spectral global correlations covering the whole HSI. To solve this drawback, graph convolutional networks (GCNs) and Transformer models equipped with self-attention mechanisms are successively introduced into this field. The two types of models can explicitly establish global context connections among different pixels [12], [13], [14]. Recently, Mamba-based state-space models (SSMs) have delivered cutting-edge performance in capturing long-range features from hyperspectral data, benefiting from their high-efficiency long-sequence modeling capacity [15], [16], [17]. Despite the great accuracy gains brought by current deep-learning-based classification approaches, most studies concentrate on optimizing feature discrimination capability and classification decision boundaries. Few works pay attention to retaining the inherent spatial topological structure of original HSI data during feature extraction, which is still an unsolved research gap in current HSI classification tasks.

Nearly all the above-reviewed classification approaches of HSI are established under the discriminative learning paradigm, whose primary objective is to directly learn the mapping relationship between input hyperspectral data and category labels. Although CNNs, Transformers, and SSMs have continuously improved feature representation capability, their underlying nature still relies on discriminative modeling based on fixed category boundaries. However, for hyperspectral data, the existence of significant within-class spectral variations, between-class spectral overlaps, along with intricate noise disturbances makes it difficult for boundary-based discriminative learning to fully characterize the intrinsic distribution structures of land-cover categories in the joint spectral–spatial space [18]. As a result, these methods are prone to boundary ambiguity and unstable predictions in complex scenarios.

From the perspective of distribution modeling, hyperspectral dense classification should essentially be regarded as a class-conditional spectral–spatial distribution restoration problem rather than a simple pixelwise discrimination task. However, existing methods generally lack the capability to explicitly model class-conditional generation mechanisms, making it difficult to achieve progressive semantic restoration of complex spectral–spatial manifolds.

Diffusion models achieve complex data distribution modeling through progressive noise perturbation and reverse denoising processes, which fundamentally constitute a progressive probabilistic distribution restoration mechanism. Compared with conventional generative models, their stable training characteristics and powerful distribution modeling capability have demonstrated significant potential in structured prediction tasks. For hyperspectral classification, diffusion models are capable of progressively recovering category semantic distributions under conditional feature guidance, thereby enabling more natural modeling of spectral–spatial distributions and category manifold structures in complex scenes. Therefore, introducing conditional diffusion mechanisms into hyperspectral dense classification is expected to overcome the reliance of traditional discriminative methods on fixed decision boundaries and facilitate a transition from boundary-based discrimination to label-space generative decoding. It should be noted that, according to the classical analysis of Ng and Jordan [19], discriminative classifiers typically outperform generative ones when training data is plentiful. Therefore, the generative formulation advocated in this work is primarily motivated by few-shot and spatially disjoint scenarios, where annotated samples are scarce, and the distribution modeling capability of generative models provides a distinct advantage.

To this end, this article introduces the hyperspectral conditional diffusion for dense (HCD^2) framework, which adapts the label-space conditional diffusion paradigm [20] to HSI dense classification with a dedicated spectral–spatial encoder, thereby enabling generative dense semantic inference. Architecturally, HCD^2 establishes a unified "HSI encoder–conditional diffusion decoder" pipeline. The encoder extracts spatial–spectral features, while the decoder reconstructs category distributions via conditional reverse diffusion. On the encoding side, we engineer an adaptive spectral decoupling (ASD) module that enforces class-consistency constraints to distinctly sharpen class-level discriminative spectral patterns. A spatial–spectral joint topological encoder (SSJTE) captures long-range spatial–spectral topologies. Ultimately, the conditional diffusion category decoder (CDCD) capitalizes on these conditional representations to iteratively generate structured category predictions, progressively refining category boundaries.

The key innovations of our work are listed as follows.

1) Adapting the label-space conditional diffusion paradigm [20] to HSI dense classification with a dedicated spectral–spatial encoder, enabling iterative semantic refinement through conditional denoising.
2) Developing a joint encoder combining an ASD module for class-consistent spectral weighting and a SSJTE for long-range spatial–spectral topological modeling.
3) Validating the framework on multiple benchmark datasets under few-shot and spatially disjoint protocols, where HCD^2 achieves competitive accuracy and stronger noise robustness over state-of-the-art methods.

II. Related Works

A. Traditional Methods of Machine Learning

Early research on classification of HSI mainly relied on pixel-level statistical analysis and artificially designed shallow spectral features to complete classification tasks. To adapt to the nonlinear data distribution inherent in raw hyperspectral datasets, scholars have applied multiple classic discriminative machine learning algorithms to HSI classification scenarios. Frequently used approaches cover multinomial logistic regression [21], Bayesian estimation algorithm [22], linear regression model [23], support vector machines [24], maximum a posteriori estimation [25], and random forest algorithm [26]. Apart from independent classification algorithms, dimensionality reduction techniques are also widely researched to tackle severe band redundancy and multicollinearity issues

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existing in high-dimensional hyperspectral data. Researchers employ sparse representation classification (SRC) [27], principal component analysis (PCA) [28], and linear discriminant analysis (LDA) to compress high-dimensional HSI data cubes, which can preserve core effective feature information while mapping original data into low-dimensional feature subspaces.

Traditional machine learning algorithms boast definite mathematical derivation processes and high model interpretability. Nevertheless, their classification performance heavily hinges on manual feature engineering and professional prior knowledge of HSI remote sensing. Such shallow learning frameworks fail to automatically excavate high-level semantic features hidden inside massive hyperspectral data. In addition, these methods cannot capture synergistic spatial–spectral correlation features effectively. Especially for complex HSI scenes with heterogeneous ground objects, the above-mentioned shallow models usually produce poor classification results with obvious misclassification pixels [29].

B. CNN-Based Methods

Driven by the vigorous progress of deep learning techniques, CNNs have evolved into the dominant network backbone for HSI classification tasks. The core superiority of CNNs lies in automatic hierarchical feature extraction, which gets rid of cumbersome manual feature design steps required by conventional machine learning methods [30]. With the support of complete end-to-end training frameworks, CNN models can adaptively learn implicit spatial texture characteristics and nonlinear spectral variation rules directly from original HSIs.

According to differences in convolution dimension settings, CNN-based HSI classification approaches can be divided into three developmental phases. Initially, 1D-CNN was designed to extract single spectral features from independent pixel spectral vectors. However, this model completely discards spatial neighborhood information of ground objects, resulting in unsatisfactory classification accuracy in practical tests [31]. After that, 2D-CNN was introduced to process spatial patch samples, where all spectral bands are treated as multichannel feature inputs consistent with ordinary RGB images. Although 2-D convolution remedies the shortage of missing spatial context information, it is incapable of describing continuous inherent correlation between adjacent spectral bands [32], [33]. To achieve synchronous fusion of spatial and spectral features, 3D-CNN adopts volumetric convolution kernels to process complete HSI data cubes. It jointly extracts spatial and spectral features from dual dimensions simultaneously, which effectively boosts overall classification performance [34].

In recent years, combining attention modules with CNNs has become a mainstream optimization direction to further improve model performance. Attention mechanisms can dynamically activate key spatial regions and sensitive spectral bands that make dominant contributions to final classification results. The integrated CNN-attention architectures have achieved competitive performance in diverse HSI application scenarios, including biomedical hyperspectral detection and analysis tasks [10], [35], [36].

Even so, CNN-driven classification methods still have two unavoidable inherent limitations. First, convolution operations are restricted by fixed local receptive fields, limiting CNNs to model long-range global spatial dependencies across the entire HSI map. Second, all CNN variants belong to typical discriminative models. During model training, these networks only focus on optimizing classification boundaries, rather than fitting the real distribution law of original hyperspectral samples. These two core limitations motivate this research to develop generative models equipped with global modeling ability and explicit data distribution fitting functions.

C. Transformer-Based Methods

To break through the local receptive field bottleneck of convolutional networks, researchers have introduced Transformer architecture into HSI classification tasks. Relying on built-in multihead self-attention mechanisms, Transformer can efficiently capture long-range global correlation relationships covering both spatial and spectral dimensions within complete HSI data [37].

Current Transformer-based HSI classification models fall into three major categories. The first category refers to the spectral Transformer, which takes each independent spectral band as sequential input data and captures subtle spectral differences among various ground materials via sequence modeling [38]. The second category is dual-branch Transformer structures, which build two separated network branches for spatial feature modeling and spectral feature modeling, respectively. This structure realizes parallel extraction of global spatial pixel correlation and interband spectral correlation, so as to further elevate classification precision [39], [40]. The last category is hybrid CNN-Transformer networks that combine the strengths of convolutional networks and Transformer. Taking SSFTT [41] as a representative work, the network adopts shallow convolution layers to extract low-level local spatial features at first, followed by Transformer encoders to realize deep fusion of global spatial and spectral features. Furthermore, token compression strategies represented by SQSFormer [42] are proposed to cut down redundant input tokens, which reduces computational overhead while retaining complete spatial–spectral correlation information.

Despite outstanding performance in global context modeling, Transformer still faces two prominent defects in HSI classification applications. For one thing, dense self-attention calculation leads to quadratic computational complexity, bringing massive computation pressure and low inference speed. For another thing, similar to convolutional models, the Transformer is also a discriminative network. When coping with complex nonlinear noise and cross-domain distribution offset existing in actual collected HSI data, the model will suffer from degraded generalization capability and robustness. Such drawbacks prompt researchers to explore lightweight sequence modeling frameworks and generative models with explicit distribution modeling functions.

D. Mamba-Based Methods

Mamba-driven selective SSMs have thrived as a novel high-efficiency sequence modeling architecture. Different from Transformer with quadratic computational complexity, Mamba

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supports linear computational complexity ($O(N)$) while maintaining powerful long-range global modeling capability. It provides a new feasible solution for high-efficiency HSI analysis [43].

Existing Mamba-based HSI classification models have been continuously optimized in network structure in recent years. The pioneering work MambaHSI [15] develops a dual-branch network structure using separate Mamba blocks to extract the spatial and spectral features, respectively, which separately captures long-range spatial pixel dependencies and fine-grained spectral variation characteristics. Compared with conventional CNN models, this dual-branch Mamba framework achieves obvious accuracy gains. Later, SS-Mamba [44] designs a shared spatial–spectral tokenizer to realize bidirectional state transmission across dual branches, strengthening interactive fusion between spatial and spectral features. Recently, hybrid compound networks combining Mamba with other mainstream backbones have become a research hotspot. MamTrans [45] alternately stacks Mamba state-space layers and Transformer layers to enhance the adaptive capacity of the model for diverse types of hyperspectral datasets. ConvMamba [46] integrates local convolution operations and Mamba sequence modeling modules, achieving a good balance between local fine-grained feature extraction and global context modeling.

Although Mamba series models have advantages of low computational cost and excellent global modeling performance, almost all existing Mamba-based HSI classification methods follow discriminative learning paradigms [47]. These models directly establish a mapping relationship between the input hyperspectral features and the output category labels, ignoring the real potential probability distribution of original HSI data. Meanwhile, these networks cannot reconstruct original HSIs or simulate the data generation process, which limits their classification performance under noisy interference and cross-domain distribution shift scenarios.

E. Generative Methods

Driven by the rapid development of generative models in computer vision fields, diffusion probabilistic models have become mature tools for fitting complex nonlinear data distributions [48], [49]. A full diffusion model operates via two key phases: noise addition(diffusion) and denoising. In the first phase, Gaussian noise is continuously injected into raw hyperspectral data in accordance with Markov chain rules. After multiple iteration steps, the distribution of original hyperspectral data will be completely converted into the standard distribution of Gaussian. In the second phase, a trainable neural network model is used to eliminate noise incrementally, recovering the true data distribution from noisy samples.

Compared with other classic generative frameworks like generative adversarial networks (GANs), the diffusion models have more stable training convergence, higher quality data generation effects, and more reliable explicit density estimation performance. Considering that most existing HSI classification methods rely on discriminative deep learning models lacking explicit distribution modeling ability, this article introduces conditional diffusion models into HSI classification tasks, aiming to remedy the inherent defects of current mainstream discriminative deep learning classification methods.

TABLE I

KEY DIFFERENCES BETWEEN HCD^2 AND RELATED DIFFUSION-BASED METHODS

Feature	DDP [20]	SpectralDiff [51]	HCD^2
Diffusion object	Encoded label map	HSI patches	Label map
Conditioning	Concatenation	None	Channel fusion + temporal modulation
Prediction target	$\mathbf{z}_0$ ($\mathbf{x}_0$-pred.)	ϵ (ϵ-pred.)	$\hat{\mathbf{z}}_0$ ($\mathbf{x}_0$-pred.)
Loss function	Task-specific CE	L_1 norm	CE + prototype alignment
Sampling	DDIM, $T=1-3$	DDPM, $T\leq 400$	DDIM, $T=3$
Encoder	Swin/ConvNeXt + FPN	3D-Conv U-Net	ASD + SSJTE
Decoder	Deformable attention	Separate attention classifier	U-Net (CDCD)
Module relation	Single pipeline	Two-stage, loosely coupled	Single pipeline
Domain	General dense prediction	HSI classification	HSI dense classification

Recent research has started to investigate the adoption of diffusion models to structured prediction tasks. DDP [20] introduced the diffusion modeling paradigm into dense visual prediction by formulating tasks such as semantic segmentation as conditional denoising processes in the output space, thereby providing a novel generative perspective for high-precision structured prediction. The label-space diffusion paradigm adopted in this work follows this formulation. As a representative generative approach, SpectralDiff [50] pioneered the application of diffusion models to HSI classification by reformulating classification as a generative denoising process and leveraging iterative evolution mechanisms to capture complex spectral distribution characteristics. FedDA-HSI [51] proposed a class-aware federated HSI classification scheme integrated with diffusion-based augmentation, where diffusion models are employed to generate class-conditioned samples for alleviating data heterogeneity and limited-sample issues in federated learning scenarios. More recently, Wang et al. [52] proposed a cross-scene diffusion-enhanced uncertainty attention network that leverages diffusion-derived features to improve HSI classification under cross-domain distribution shifts; Zhu and Xu [53] introduced a semi-supervised conditional diffusion model for language-informed HSI synthesis to address imbalanced small-sample classification. These methods primarily leverage diffusion models for feature enhancement or data synthesis, which differs fundamentally from HCD^2 in that our framework reformulates the classification task itself as a conditional label-map denoising process, enabling generative dense semantic inference rather than serving as an auxiliary enhancement or synthesis module.

Table I summarizes the key differences between HCD^2 and the two most related diffusion-based methods. DDP [20] first introduced the label-space diffusion paradigm for general dense prediction, where diffusion operates on the encoded

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label map conditioned on image features from a generic backbone. HCD^2 adopts this label-space formulation but differs in two aspects: 1) a dedicated HSI encoder (ASD + SSJTE) with spectral attention and prototype alignment is designed instead of a generic backbone; and 2) a U-Net-based decoder (CDCD) is adopted for multiscale semantic restoration. SpectralDiff [50] takes a fundamentally different approach: it applies unconditional diffusion to HSI spectral–spatial patches using ϵ-prediction, and employs the U-Net denoising network as a feature extractor whose intermediate activations are fed to a separate attention-based classifier. The reverse sampling is not used for classification. Consequently, SpectralDiff leverages diffusion as unsupervised feature learning, whereas HCD^2 directly generates the label distribution via conditional iterative denoising at the output level.

Nevertheless, diffusion models still face unique deployment barriers when applied to practical HSI analysis tasks, which stem from the inherent structural characteristics of hyperspectral tensor data. Specifically, hyperspectral data carries massive high-dimensional spectral collinearity and redundant band information, and conventional diffusion models adopt sample-focused localized attention strategies, which cannot effectively capture overall category-level spectral structural features across the whole dataset. More importantly, although mainstream discriminative deep learning models can build accurate mapping between extracted features and ground object labels, they ignore essential underlying data distribution modeling. Such defects make these models vulnerable to complex spectral variations and random noise widely existing in real-scene HSI data. In contrast, other existing generative models also have obvious limitations: they are hard to reconstruct complete dense spatial–spectral continuous manifolds of HSI data, resulting in poor fine-grained classification performance under limited labeled sample conditions.

Targeting the above coupled technical bottlenecks of discriminative models and traditional generative models in HSI classification, this study proposes a novel HCD^2 network framework. Combined with conditional diffusion mechanisms, the HCD^2 model integrates explicit generative data distribution learning and discriminative feature optimization modules organically. This dual collaborative design not only enhances the models capability of extracting refined local spatial–spectral features but also improves the overall spatial generalization ability of the network for complex remote sensing scenes.

III. Method

A. Dense Prediction for Conditional Diffusion

Formally, HSI classification is formulated as a pixelwise dense semantic inference task. Given an input hyperspectral tensor $\mathbf{x} \in \mathbb{R}^{B\times H\times W}$, where B denotes the continuous spectral bands' number and $H \times W$ represents the spatial dimensions, the objective is to map each pixel vector to its corresponding land-cover category within a predefined label space $\mathcal{Y} = \{1, 2, \ldots, N\}$. Here, N explicitly designates the total number of distinct thematic classes. Accordingly, the target dense prediction output is defined as a structured label matrix

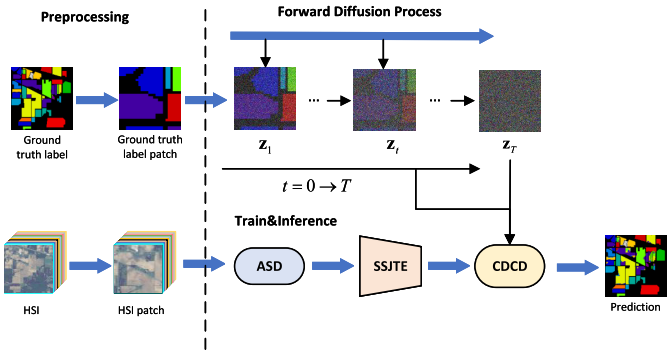


Fig. 1. Framework of HCD^2.

$\mathbf{y} \in \mathcal{Y}^{H\times W}$, where each element indexes a specific thematic land-cover category.

Different from conventional discriminative classification methods, this article reformulates HSI classification as a conditional diffusion-driven dense prediction problem. Specifically, under the guidance of hyperspectral conditional information, the diffusion model progressively generates pixelwise category distribution maps from random noise.

As illustrated in Fig. 1, the diffusion process in the proposed framework is performed on the category label map rather than the input HSI itself. Let the ground-truth label map be denoted as $\mathbf{z}_0 = \mathbf{y}$. The discrete category labels are first transformed into continuous representations through one-hot encoding followed by linear scaling to the interval $[-s, s]$, so as to satisfy the continuous-data modeling assumption of diffusion models. Formally, for a pixel with category k, the one-hot vector $\mathbf{e}_k \in \{0, 1\}^N$ is linearly scaled as $\mathbf{z}_0 = (2\mathbf{e}_k - \mathbf{1}) \cdot s \in [-s, s]^N$, placing the clean label at the vertices of a hypercube in the continuous latent space. This continuous relaxation enables the application of standard Gaussian diffusion transitions while preserving the categorical structure through the vertex separation.

The forward diffusion process progressively injects Gaussian noise into the label representation, transforming $\mathbf{z}_0$ into a noisy label map $\mathbf{z}_t$, which is formulated as

$$q(\mathbf{z}_t \mid \mathbf{z}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{z}_0,\ (1 - \bar{\alpha}_t)\mathbf{I}\right) \tag{1}$$

where

$$\bar{\alpha}_t = \prod_{s=1}^{t} (1 - \beta_s) \tag{2}$$

and β_s denotes the noise scheduling parameter. In this work, a cosine noise scheduling strategy is adopted to achieve a more stable signal-to-noise ratio evolution process.

During the training stage, the model learns a conditional denoising network $f_\theta(\mathbf{z}_t, \mathbf{x}, t)$, which predicts the corresponding clean label representation $\mathbf{z}_0$ from the noisy label map $\mathbf{z}_t$ under the condition of the input HSI $\mathbf{x}$.

During inference, the model starts from random Gaussian noise $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and progressively generates the final prediction through a multistep Markov reverse diffusion process

$$p_\theta(\mathbf{z}_0 \mid \mathbf{x}) = \prod_{t=1}^{T} p_\theta(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}). \tag{3}$$

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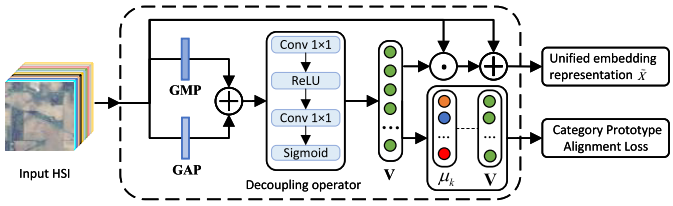


Fig. 2. Illustration of the proposed ASD module, which performs category-aware spectral weight learning to capture discriminative spectral structures.

B. Adaptive Spectral Decoupling

HSI contains extremely rich spectral information across high-dimensional spectral bands. However, the strong inter-band correlations lead to severe information redundancy and the curse of dimensionality, which constitute major bottlenecks restricting classification performance.

To address the problem, an ASD module is designed, as illustrated in Fig. 2. The core idea is to apply category-aware spectral attention to adaptively weight discriminative spectral bands, suppress redundant components with low contributions, and construct a stable spectral–spatial unified embedding. This design builds upon the channel attention paradigm [54], [55], extending it to the spectral dimension of hyperspectral data and coupling it with a category prototype alignment constraint to enforce class-consistent spectral weighting within the diffusion pipeline.

Given an input hyperspectral tensor $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$, the ASD module first extracts spectral discriminators through a spatial topological aggregation operator. Different from conventional methods that rely on a single statistical descriptor, a parallel dual-branch structure is adopted in this work: saliency extraction branch: global max pooling is employed to capture instantaneous strong activation signals in spectral responses; Energy distribution branch: global average pooling is introduced to characterize the overall energy distribution of spectral bands.

The spectral weighting operator derives the control variables in the spectral space as follows:

$$V = \sigma\left(F_{1\times1}\left[\delta\left(F_{1\times1}\left(\mathrm{GAP}\left(X\right)+\mathrm{GMP}\left(X\right)\right)\right)\right)\right) \tag{4}$$

where $\sigma(\cdot)$ and $\delta(\cdot)$ denote the Sigmoid and ReLU activation functions, respectively.

To ensure that the decoupled features maintain physical consistency within the same semantic category, a category prototype alignment loss is introduced. Specifically, we assume that the optimal weighting state should drive the spectral control vectors of samples belonging to the same category toward the corresponding class-specific "feature center." Let μ_k be the decoupling center (prototype) of the kth category. For the ith sample, the consistency constraint is defined as

$$\mathcal{L}_{\mathrm{align}} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{V}_i - \boldsymbol{\mu}_{y_i}\right\|_2^2. \tag{5}$$

The operational mechanics of the category prototypes $\{\boldsymbol{\mu}_k\}_{k=1}^{N}$ are detailed as follows.

Initialization: Before training, a forward pass is performed on the entire training set using the randomly initialized network. The spectral control vectors $\mathbf{V}_i$ of all samples are extracted and averaged per category to obtain the initial prototypes $\boldsymbol{\mu}_k^{(0)}$, providing a stable starting point that reflects the coarse spectral structure of each class.

Update Strategy: The prototypes are not learnable parameters and receive no gradient during backpropagation. Instead, they are updated via exponential moving average (EMA) after each training epoch

$$\boldsymbol{\mu}_k \leftarrow (1-\eta)\,\boldsymbol{\mu}_k + \eta\,\frac{1}{|\mathcal{S}_k|}\sum_{i\in\mathcal{S}_k}\mathbf{V}_i \tag{6}$$

where $\mathcal{S}_k$ denotes the set of samples belonging to category k in the current epoch and $\eta = 0.1$ is the EMA decay rate. This momentum-based update tracks the slow evolution of the feature space without being perturbed by per-batch fluctuations. Warm-up mechanism: During the first $E_w = 10$ epochs, the feature space is still highly chaotic, and enforcing strict alignment prematurely may lead to feature collapse or suboptimal local minima. Therefore, the alignment loss $\mathcal{L}_{\mathrm{align}}$ is disabled (i.e., $\lambda = 0$) for the first E_w epochs and linearly ramped up to its target value over the subsequent 5 epochs, allowing the encoder to first establish a reasonably structured feature manifold before the prototype-based regularization takes effect.

Finally, the decoupled spectral features are fused with the original spatial information through a residual mapping strategy to generate the unified embedding representation

$$\tilde{X} = X \odot V + X. \tag{7}$$

C. Spatial–Spectral Joint Topological Encoder

After category-level spectral decoupling in the ASD module, the resulting hyperspectral embedding $\tilde{X}$ is fed into the SSJTE to construct a spectral–spatial joint manifold representation for conditional diffusion restoration. Different from conventional encoders that mainly focus on local discriminative feature extraction, this article considers that hyperspectral land-cover categories are inherently distributed over complex spectral–spatial topological manifolds, where spatial structures and spectral patterns exhibit significant variations across different scales. For instance, elongated targets such as roads and rivers rely heavily on local continuity structures, whereas farmland and vegetation regions depend more on large-scale contextual consistency. Therefore, serial convolutional architectures with fixed receptive fields are insufficient for comprehensively modeling the complex multiscale topological relationships in practical scenes.

Aiming to solve this issue, a joint encoding mechanism based on multiscale feature synergy is constructed in this work. Specifically, dynamic spectral compression (DSC) and multiscale spatial topology extraction (MSTE) are collaboratively designed to jointly model spectral–spatial structural information.

First, to reduce high-dimensional spectral redundancy and enhance cross-band semantic consistency, an adaptive 1×1 convolution is employed to perform nonlinear projection on the decoupled high-dimensional spectral embedding

$$F_{\mathrm{spec}} = F_{1\times1}\left(\tilde{X}\right). \tag{8}$$

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This process not only compresses the spectral dimensionality but also aggregates cross-band information through parameter sharing, thereby mapping discrete spectral responses into a more discriminative latent spectral manifold space.

To fortify the model's robustness against complex spatial topographies, a parallel dual-receptive feature extraction network is implemented to map multiscale spatial representations. Formally, given the antecedent spectral embedding $\mathbf{F}_{\text{spec}}$, the local and contextual streams are defined as follows:

$$F_{\text{local}} = \mathcal{K}1\left(F_{\text{spec}}\right), \quad F_{\text{context}} = \mathcal{K}3\left(F_{\text{spec}}\right) \tag{9}$$

where $\mathcal{K}_1$ symbolizes a standard 3×3 convolution optimized for fine-grained boundary extraction, while $\mathcal{K}_3$ denotes a dilated convolution with a dilation factor of 2, expanding the perceptual fields to route long-range, nonlocal spatial interactions.

To further enhance the collaborative representation capability among multiscale spectral–spatial features, an interactive gating unit (IGU) is introduced to achieve dynamic selection and nonlinear fusion of multiscale structural information

$$C = \text{Concat}\left[F_{\text{local}}, F_{\text{context}}\right] \otimes \sigma\left(F_{1\times 1}\left(F_{\text{spec}}\right)\right). \tag{10}$$

The final conditional feature map $C \in \mathbb{R}^{H\times W\times d}$ not only contains rich local spatial textures and global contextual structures, but also encodes category-related spectral–spatial topological priors.

D. Conditional Diffusion Category Decoder

In conventional discriminative classification frameworks, category prediction is typically accomplished through a single forward mapping process. However, for HSIs, the presence of complex noise interference, mixed pixels, and significant intraclass spectral variability makes it difficult to stably recover category semantic structures in complex scenes using only one-step discriminative mapping. To address this issue, the proposed CDCD reformulates hyperspectral dense classification as a progressively conditioned semantic distribution restoration process. The rationale is that HSI features provide a stable conditional prior that guides the label-map denoising trajectory, while iterative denoising progressively refines category boundaries and suppresses noise-induced misclassification. At each reverse step, the network performs a soft classification conditioned on the spectral–spatial features while simultaneously enforcing neighborhood consistency in the label space, enabling gradual correction of within-class spectral variations and noise artifacts that single-pass discriminative mapping cannot resolve.

As illustrated in Fig. 3, at diffusion step t, the decoder receives three types of conditional inputs: the noisy label state $\mathbf{z}_t$, the spectral–spatial conditional feature map C which is generated by the encoder, and the temporal embedding vector. Here, $\mathbf{z}_t$ represents the category distribution state at the current diffusion stage, while the conditional feature map C encodes the spectral–spatial topological priors contained in the HSI.

To continuously maintain category semantic consistency during the reverse diffusion process, a condition-guided mechanism is adopted in this work, where the static spectral–spatial

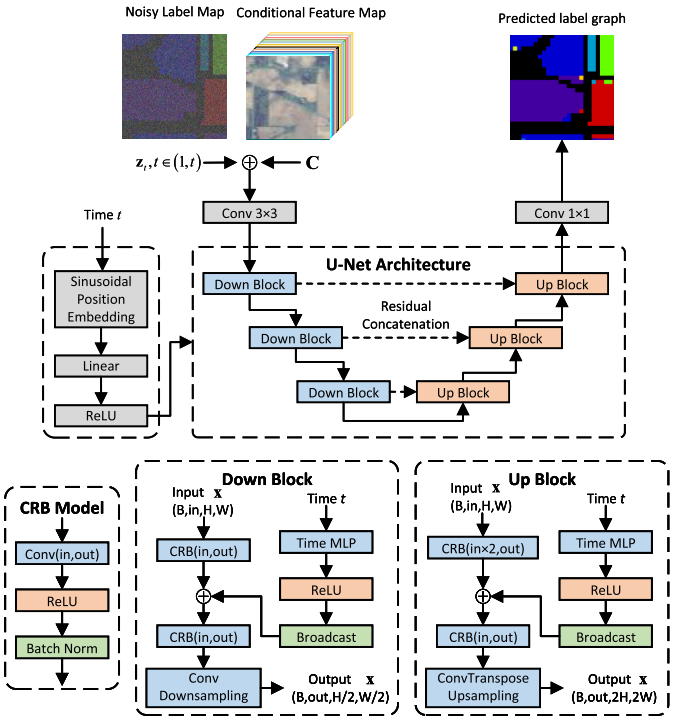


Fig. 3. Illustration of the proposed CDCD module.

conditional features and the dynamic noisy states are fused along the dimension of channels. As a result, the model can consistently capture the spatial structures and spectral attributes of ground objects throughout each reverse diffusion step, thereby alleviating the gradual degradation of conditional information typical of conventional conditional diffusion models. Furthermore, it constrains the reverse diffusion trajectory to converge toward the real category manifold, enabling stable restoration of conditional semantic distributions.

Simultaneously, a temporal embedding module based on sinusoidal positional encodings injects time-conditioned features into each residual block, executing dynamic scaling across distinct degradation phases. This time-step modulation adaptively recalibrates the semantic restoration trajectory conditioned on the instantaneous noise variance, explicitly stabilizing the inverse diffusion trajectory while enhancing the spatial coherence of multiclass semantic boundaries.

In terms of overall architecture, the proposed CDCD adopts a lightweight U-Net structure to achieve multiscale semantic restoration. Skip connections are utilized to fuse shallow spatial details with deep semantic information, thereby enabling high-precision boundary recovery in complex scenarios.

Different from conventional diffusion models that adopt the ϵ-prediction parameterization (i.e., predicting the injected noise), the proposed framework adopts the $\mathbf{x}_0$-prediction parameterization, in which the denoising network directly predicts the original category distribution $\hat{\mathbf{z}}_0$. As discussed in [56], the $\mathbf{x}_0$-prediction and ϵ-prediction parameterizations are mathematically equivalent up to a rescaling factor; however, the former is more natural for structured prediction tasks where the target is a categorical distribution, since it avoids the additional conversion step from noise to data and allows direct

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supervision on the recovered label representation. Through iterative conditional reverse diffusion, the model progressively refines category predictions and restores spatially continuous structured category distributions, ultimately generating high-precision dense classification results in the original HSI space.

E. Task-Driven Training and Inference Strategy

In training process, the model aims to reduce prediction error during reverse denoising. Specifically, the ground-truth label map $\mathbf{y}_0$ is first mapped into the noisy space through the forward diffusion operator q, yielding the noisy observation $\mathbf{y}_t$ at a randomly sampled time step t. Based on this approach, the model can learn to progressively recover the original label distribution from the noisy state under the guidance of the hyperspectral conditional feature C.

To enhance both discriminative capability and generative quality, a hybrid task-driven supervision objective is constructed in this work, jointly constraining the diffusion denoising accuracy and the spectral structure consistency. The total loss function may be formulated as

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda \mathcal{L}_{\text{align}} \tag{11}$$

where $\mathcal{L}_{\text{diff}}$ denotes the semantic-space denoising loss. Conventional diffusion models typically adopt the ϵ-prediction parameterization, where the network is trained to predict the injected Gaussian noise ϵ and the loss is formulated as mean squared error in the continuous latent space [56]. However, for dense classification tasks, the ultimate objective is to recover the clean category distribution rather than the noise itself. Therefore, this work adopts the $\mathbf{x}_0$-prediction parameterization (also referred to as data prediction), in which the conditional denoising network f_θ directly predicts the clean label representation $\hat{\mathbf{z}}_0$ from the noisy state $\mathbf{z}_t$. The network output is passed through a softmax function to produce a valid probability distribution over the N land-cover categories, and cross-entropy loss is computed against the ground-truth one-hot encoded label $\mathbf{z}_0$

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{(\mathbf{z}_0,\epsilon,t)}\left[-\sum_{i=1}^{H\times W} \mathbf{z}_{0,i} \log\left(\text{softmax}\left(f_\theta\left(\mathbf{z}_t, t, \mathbf{C}\right)\right)_i\right)\right] \tag{12}$$

where $f_\theta(\mathbf{z}_t, t, \mathbf{C})$ denotes the conditional denoising network that directly predicts the clean label distribution from the noisy state $\mathbf{z}_t$ at diffusion step t, conditioned on the spectral–spatial feature map $\mathbf{C}$. Here, f_θ corresponds to the denoising network $f_\theta(\mathbf{z}_t, \mathbf{x}, t)$ introduced in Section III-A, consistently parameterized under the $\mathbf{x}_0$-prediction scheme rather than the ϵ-prediction scheme. The expectation is taken over the clean label $\mathbf{z}_0$, the injected noise ϵ (which determines $\mathbf{z}_t$ through the forward process), and the time step t.

Although the forward process operates in the continuous latent space via Gaussian transitions [see (1)] and the loss is formulated as cross entropy on a discrete probability simplex [see (12)], the two are mathematically compatible through the following bridge. The network output $\mathbf{f}_\theta$ produces logits that are mapped to a valid categorical distribution via softmax, i.e., $\hat{\mathbf{p}} = \text{softmax}(\mathbf{f}_\theta) \in \Delta^{N-1}$, where Δ^{N-1} denotes the $(N-1)$-dimensional probability simplex. The cross-entropy loss $-\mathbf{z}_0^\top \log \hat{\mathbf{p}}$ is equivalent to the Kullback–Leibler divergence $\text{KL}(\mathbf{z}_0 \| \hat{\mathbf{p}})$ up to a constant, which constitutes the negative log-likelihood term in the variational lower bound of the reverse process. In standard DDPM [56], this ELBO term reduces to mse on the predicted $\hat{\mathbf{z}}_0$ when the data is continuous. In our formulation, since the clean target $\mathbf{z}_0$ resides at the vertices of the scaled one-hot hypercube and the network output is projected onto the simplex via softmax, the cross entropy serves as the natural categorical analog of mse: both minimize the discrepancy between the predicted and target distributions, but CE operates on the probability simplex while mse operates in the raw continuous space. This design is consistent with the spirit of discrete diffusion frameworks such as D3PM [57], which establish that categorical cross entropy provides a valid variational objective for diffusion processes targeting discrete distributions. The continuous Gaussian forward process acts as a smooth noise injection mechanism on the relaxed label representation, while the softmax-constrained CE loss ensures that the reverse process converges to valid category distributions.

Meanwhile, $\mathcal{L}_{\text{align}}$ denotes the category prototype alignment loss introduced in the ASD module, which is employed to constrain the consistency of spectral weights.

In the model inference phase, the label map $\mathbf{z}_T$ is initialized via a random Gaussian noise distribution, and the whole reverse diffusion workflow is conducted under the constraint of extracted hyperspectral conditional features. To strike a reasonable balance between generated label map quality and real-time inference speed, this work adopts the denoising diffusion implicit models (DDIMs) sampling scheme. Different from conventional uniform time step settings, this method applies nonuniform time step sampling to cut down redundant iteration steps and speed up model convergence during inference.

It is worth noting that the number of sampling steps required by the proposed framework is far smaller than that of image-generation diffusion models. This stems from the fundamentally simpler structure of the target distribution: the clean label maps reside near the vertices of the scaled one-hot hypercube, so the reverse process essentially needs to route each pixel toward one of N category prototypes under the guidance of the conditional features, rather than reconstruct high-dimensional continuous textures. A similar few-step saturation has also been observed in dense prediction diffusion models [20]. Moreover, the benefit of the diffusion formulation is not confined to multistep inference: during training, the network is supervised to recover the complete label distribution at every noise level, which imposes a stronger regularization than the single-level discriminative objective and shapes the conditional features accordingly, whereas at inference time only a few DDIM steps are needed to materialize this learned recovery capability.

After the multistep DDIM sampling concludes, the final estimated continuous tensor $\hat{\mathbf{z}}_0 \in \mathbb{R}^{H\times W\times N}$ is mapped back to the discrete label space by applying argmax along the channel dimension: $\hat{\mathbf{y}} = \text{argmax}_N(\hat{\mathbf{z}}_0)$, yielding the final dense classification map. Since the argmax operation is scale-invariant, the initial scaling parameter s does not affect the

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final discrete decision; it only controls the separation between category representations in the continuous latent space during the diffusion process, which influences the smoothness of the reverse trajectory but not the final label assignment.

Benefiting from the inherent iterative optimization property of diffusion models, the developed HCD^2 framework supports flexible dynamic inference paths. In practical testing, researchers can adjust the number of sampling steps freely, so as to adaptively balance computational inference cost and final classification mapping accuracy according to actual task demands.

IV. Experimental Setup and Results Analysis

A series of comparative experiments are carried out in this section to fully verify the classification accuracy and cross-scene generalization performance of HCD^2 framework under diverse complex remote sensing topological environments. Two separate evaluation schemes are designed in the experimental system for comprehensive performance verification. The first one is random few-shot training setting, which aims to test the feature learning capacity of the model under the condition of limited annotated training samples. The second one adopts spatially separated training and validation datasets; this scheme eliminates the interference brought by spatial autocorrelation of HSI pixels, so that the model's out-of-domain generalization ability may be evaluated more objectively and accurately.

Three widely used public hyperspectral datasets collected by different imaging sensors are selected for quantitative and qualitative experiments, covering diverse ground object distribution scenarios. The detailed dataset information is listed as follows.

1) Indian Pines (IP) dataset, with a 145 $\times$ 145 spatial size of pixels, 200 valid spectral bands, and 16 ground object categories. This airborne hyperspectral dataset features severe spectral similarity among different ground objects and obvious sample imbalance across categories, posing great challenges to classification models.
2) Pavia University (PU) dataset, covering 610 $\times$ 340 pixels, 103 spectral bands with nine object categories. This urban scene dataset contains abundant fine spatial texture details, which is suitable for evaluating the model's performance on accurate segmentation of intricate ground object boundaries.
3) WHU-Hi-HongHu (HH) dataset, with a 949 $\times$ 1211 spatial size of pixels, 270 spectral bands, and 22 land cover classes. Acquired by an uncrewed aerial vehicle (UAV) with an ultrahigh subdecimeter spatial resolution of 0.043 m, covering complicated crop distribution patterns and obvious intraclass spectral variations. It can serve as a challenging test case to verify the structural robustness of different classification models.

All experimental codes and deep learning modules were built based on the PyTorch framework, the experiment platform was a desktop workstation powered by Intel i9-11900F and NVIDIA RTX4060 Ti. For training the proposed HCD^2, we adopted the Adam optimizer, and the initial learning rate was set to 0.001. We also adopted cosine annealing to dynamically adjust the learning rate during 100 epochs, and set the batch size to 32 across all comparative models. For the conditional diffusion module embedded in the proposed HCD^2 framework, the forward diffusion process was set with 1000 time steps. Meanwhile, the accelerated DDIM sampling strategy was adopted in the reverse denoising process to reduce computational latency during model inference. Overall accuracy (OA), average accuracy (AA) and Kappa coefficient were utilized as core evaluation criteria for quantitative analysis. To assess the statistical stability of the reported results, all comparison and ablation experiments were independently repeated ten times with different random seeds, and the accuracy metrics are reported as the mean and standard deviation across the ten runs. We conducted extensive comparisons with a series of advanced algorithms to verify the advantages of our presented method. The test group includes CNN-related networks, Transformer structures, Mamba-enabled SSMs and mainstream generative classification approaches. All competing methods retained the architecture variants, patch sizes, and optimizers recommended in their original publications, and no hyperparameters other than the unified batch size were tuned per baseline.

TABLE II
Sample Distribution Statistics

Dataset	Train	Test
Indian Pines (IP)	444	9805
Pavia University (PU)	270	42506
WHU-Hi-HongHu (HH)	660	386033

A. Randomized Few-Shot Experiments and Result Analysis

In this few-shot classification experiment, limited labeled pixels were randomly selected from each land cover category of the three datasets to construct training sets. In general, 30 labeled samples were chosen for each category under uniform experimental settings. For a small number of minority classes with insufficient total samples, where the absolute available sample pool fell below 30, we adopted a 1:1 train–test segmentation ratio to avoid model overfitting in the training phase. The specific sample numbers for each category are presented in Table II.

It should be noted that, under this random sampling protocol, the selected training pixels are used within spatial patches that serve as the inputs to the proposed model. Therefore, a training patch may contain neighboring pixels belonging to the test set, and the spectral and spatial information of these pixels may consequently be included in the training input. Although the labels of these test pixels are not used for model optimization, this spatial contextual overlap may introduce a certain degree of spatial autocorrelation and potentially lead to optimistic classification performance. Such patch-based random sampling is commonly adopted in HSI classification, where local spatial context plays an important role in spectral–spatial feature extraction [41], [58]. To further assess the influence of spatial autocorrelation, we additionally conduct spatially disjoint experiments in Section IV-B, where

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TABLE III

QUANTITATIVE CLASSIFICATION COMPARISON ON THE IP DATASET (30 TRAINING SAMPLES PER CLASS). RESULTS ARE REPORTED AS THE MEAN AND STANDARD DEVIATION OVER TEN INDEPENDENT RUNS

Class No.	SSGRN[60]	SSRN[59]	SSFTT[42]	SQSFormer[43]	MambaHSI[15]	ConvMamba[47]	SpectralDiff[51]	Ours
1	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.75±2.50	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
2	86.41±3.92	87.18±4.16	90.21±3.36	84.18±8.06	90.51±4.04	**94.84±2.45**	85.48±4.71	89.74±3.02
3	92.38±2.87	87.80±4.17	91.60±2.50	93.03±3.15	89.19±4.18	93.23±2.20	94.13±2.36	**94.51±1.94**
4	**100.00±0.00**	99.61±0.56	99.13±1.74	96.33±1.72	96.95±4.19	99.52±0.68	**100.00±0.00**	**100.00±0.00**
5	95.14±2.73	94.75±3.24	91.08±3.97	93.16±3.46	93.05±0.99	95.50±2.50	**96.03±2.11**	91.49±3.18
6	98.14±1.12	98.00±1.01	97.06±2.04	99.54±0.39	98.24±1.25	98.16±1.26	99.86±0.29	**100.00±0.00**
7	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	**100.00±0.00**	99.82±0.36	99.73±0.54	**100.00±0.00**	99.95±0.10	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
9	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.00±4.00	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	86.84±3.41	90.51±2.65	92.27±1.83	92.95±2.52	92.65±2.97	93.88±2.74	84.50±4.37	**95.31±2.06**
11	90.68±3.76	86.32±3.17	79.69±4.43	90.01±5.56	88.75±2.62	92.76±2.44	91.96±3.58	**94.71±2.17**
12	95.20±2.31	94.07±2.44	89.73±2.48	**96.23±2.20**	92.80±2.45	94.92±2.75	95.56±2.08	93.82±2.63
13	**100.00±0.00**	98.74±1.96	98.63±1.38	99.54±0.56	**100.00±0.00**	99.29±1.51	**100.00±0.00**	**100.00±0.00**
14	92.79±2.63	97.20±1.99	94.90±4.66	98.62±1.16	97.89±0.76	97.99±2.81	97.09±2.94	**99.91±0.23**
15	**100.00±0.00**	98.71±1.86	99.16±0.73	99.61±0.38	99.51±0.69	99.39±0.94	**100.00±0.00**	**100.00±0.00**
16	**100.00±0.00**	99.05±1.27	98.41±1.00	**100.00±0.00**	98.79±1.48	99.60±0.69	98.41±2.03	**100.00±0.00**
OA(%)	92.34±1.34	91.70±1.06	90.18±1.68	93.19±1.50	92.94±1.00	95.40±0.52	93.15±1.41	**95.65±0.47**
AA(%)	96.10±0.78	95.73±0.61	95.10±0.96	96.25±0.64	96.14±0.69	97.44±0.29	96.44±0.73	**97.46±0.31**
$\kappa \times 100$	91.25±1.51	90.56±1.20	88.84±1.89	92.22±1.70	91.90±1.30	94.74±0.59	92.16±1.58	**95.05±0.43**

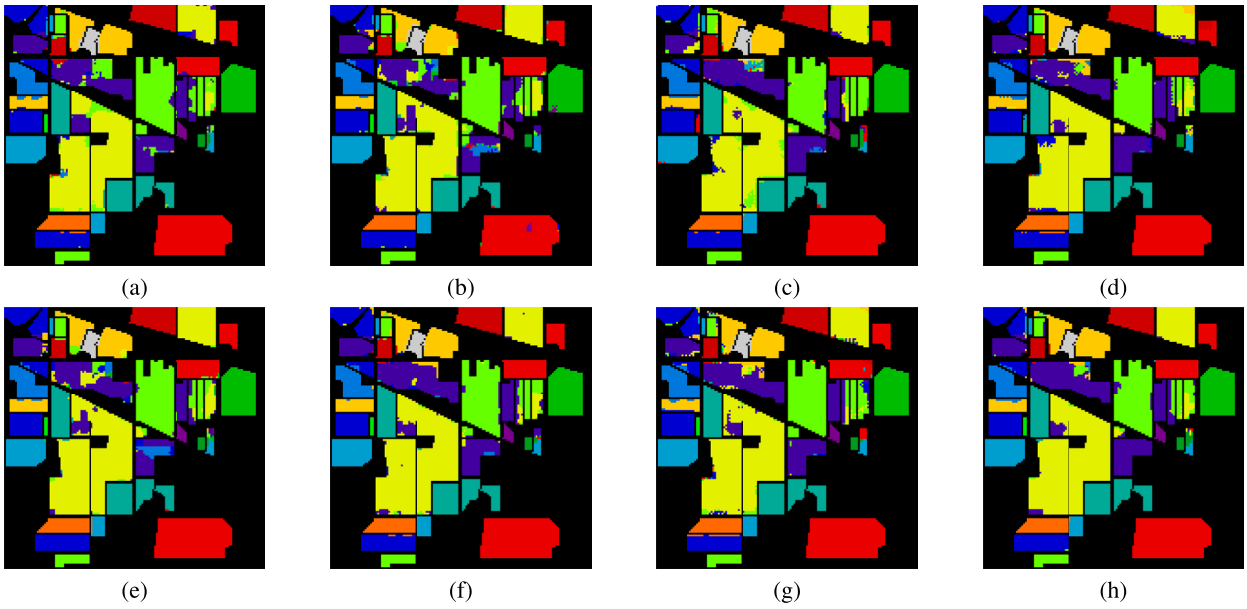


Fig. 4. Visual classification results with other methods on the IP dataset. (a) SSGRN. (b) SSRN. (c) SSFTT. (d) SQSFormer. (e) MambaHSI. (f) ConvMamba. (g) SpectralDiff. (h) Ours.

the training and testing samples are spatially separated at the patch level, providing a more rigorous evaluation of the model's spatial generalization capability.

Under the randomized few-shot protocol, corresponding quantitative classification results of all comparison methods on IP, PU, and HH datasets are listed in Tables III–V, respectively.

The results across three benchmarks demonstrate the overall superiority of HCD^2. Specifically, on the IP matrix, HCD^2 yields a peak OA of 95.65% ± 0.47%, significantly outperforming traditional CNN-centric networks like SSRN (91.70% ± 1.06%) and Transformer-based frameworks such as SSFTT (90.18% ± 1.68%), while slightly outstripping the contemporary ConvMamba baseline (95.40% ± 0.52%). For the highly fragmented urban topographies of the PU scene, the proposed framework achieves a new state-of-the-art on all main diagnostic axes (OA = 98.28% ± 0.58%, AA = 98.05% ± 0.44%, and κ = 97.72% ± 0.66%), verifying its superior fidelity in routing joint spectral–spatial topologies under severe small-sample constraints. Crucially, when scaled to the massive, high-resolution HH benchmark (over 380 000 verification pixels), HCD^2 achieves the best accuracy (OA = 95.59% ± 0.51%). Notably, while advanced selective SSMs (Mamba) thrive under data scarcity, their structural scanning boundaries degrade under the intense spatial heterogeneity of the HH scene. In contrast, HCD^2 mitigates this through conditional iterative denoising, demonstrating robustness against complex distributions.

Visual benchmarking via the qualitative classification maps in Figs. 4–6 confirms that HCD^2 generates the most spatially coherent and topologically robust land-cover segmentations.

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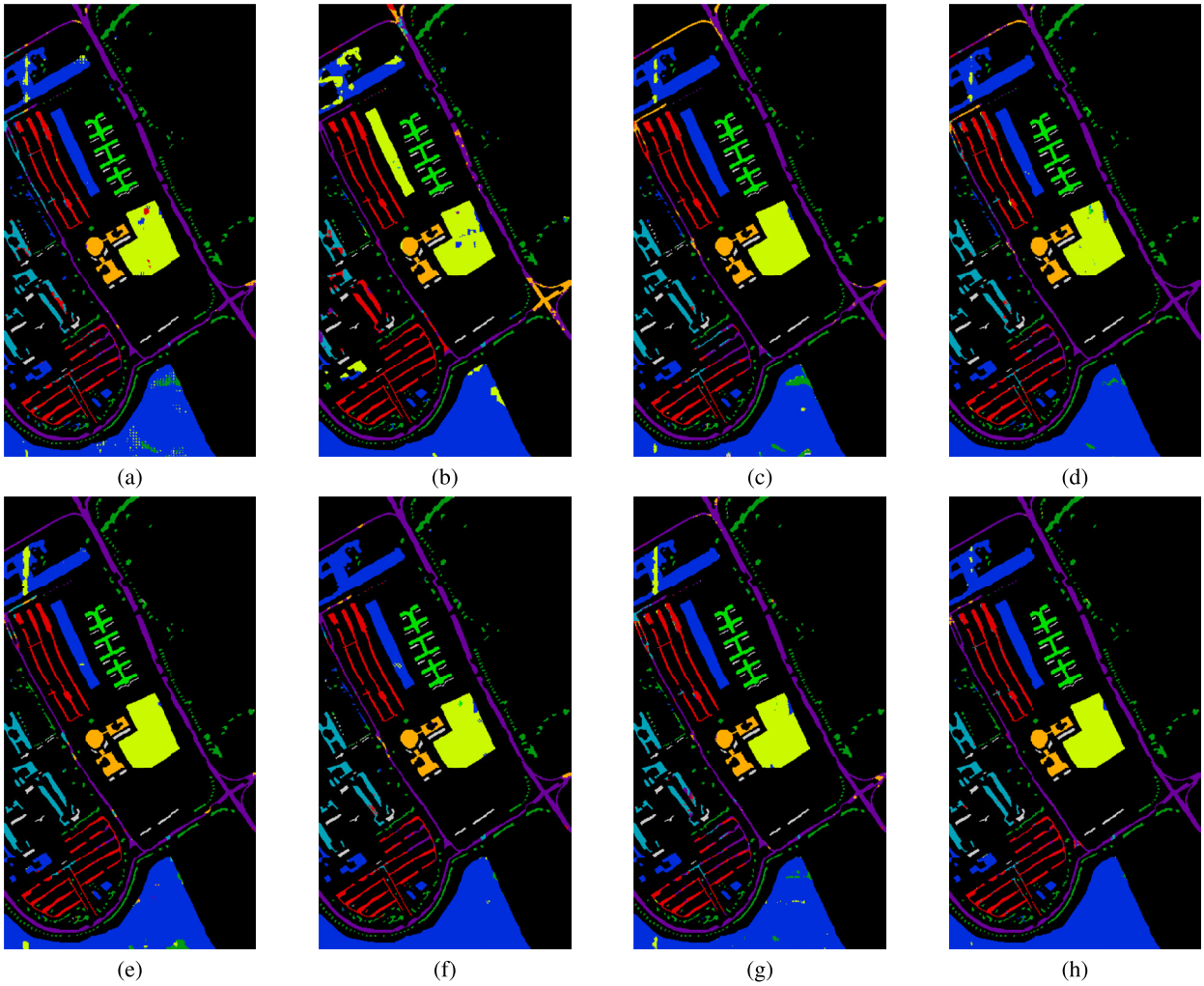


Fig. 5. Visual classification results with other methods on the PU dataset. (a) SSGRN. (b) SSRN. (c) SSFTT. (d) SQSFormer. (e) MambaHSI. (f) ConvMamba. (g) SpectralDiff. (h) Ours.

While prototypical discriminative models (e.g., SSRN and SSFTT) exhibit high vulnerability to spectral overfitting when constrained by low-shot annotations—culminating in fragmented salt-and-pepper disruptions and stochastic pixel misclassifications across continuous fields—HCD^2 preserves severe spatial-thematic consistency. Relying on iterative denoising characteristics of the diffusion structure, the HCD^2 model can explicitly mine combined spatial and category correlations from surrounding pixel neighborhoods. In this way, the model achieves automatic generative bias correction throughout the inference process without extra manual post-processing operations.

Apart from standard comparative experiments, we further test the robustness of HCD^2 facing complex practical interferences. In this part, multiple real HSI degradation scenarios are simulated, which are commonly caused by airborne platform jitter and long-term sensor hardware aging in actual remote sensing data acquisition missions. Specifically, controlled stress-testing scenarios were generated by superimposing additive Gaussian white noise across distinct intensity gradients ($\sigma \in \{0.02, 0.04, 0.06, 0.08\}$) onto the verification partition of HH. All competitive models were trained strictly on pristine hyperdimensional profiles and cross-evaluated on these corrupted observations without fine-tuning, thereby revealing their latent noise-immune capabilities across diverse architectural paradigms, including the discriminative baselines (CNN, Transformer, and Mamba) and the diffusion-based SpectralDiff [50].

As shown in Fig. 7, all the methods exhibit different levels of accuracy decline as the noise intensity increases. The discriminative models (e.g., SSRN, SQSFormer, and ConvMamba) suffer the most severe degradation, whereas the two diffusion-based methods degrade more gracefully, confirming the noise-suppression benefit of the diffusion mechanism itself. Meanwhile, HCD^2 consistently outperforms SpectralDiff across all noise levels, and the performance gap widens as the noise intensity increases. This indicates that, beyond the shared diffusion paradigm, the dedicated ASD + SSJTE encoder further contributes to the robustness: the category-aware spectral weighting suppresses noise-corrupted redundant bands, and the long-range spatial topological priors stabilize the conditional features under strong interference. These results verify that the noise robustness of HCD^2 stems jointly from the diffusion mechanism and the proposed encoder.

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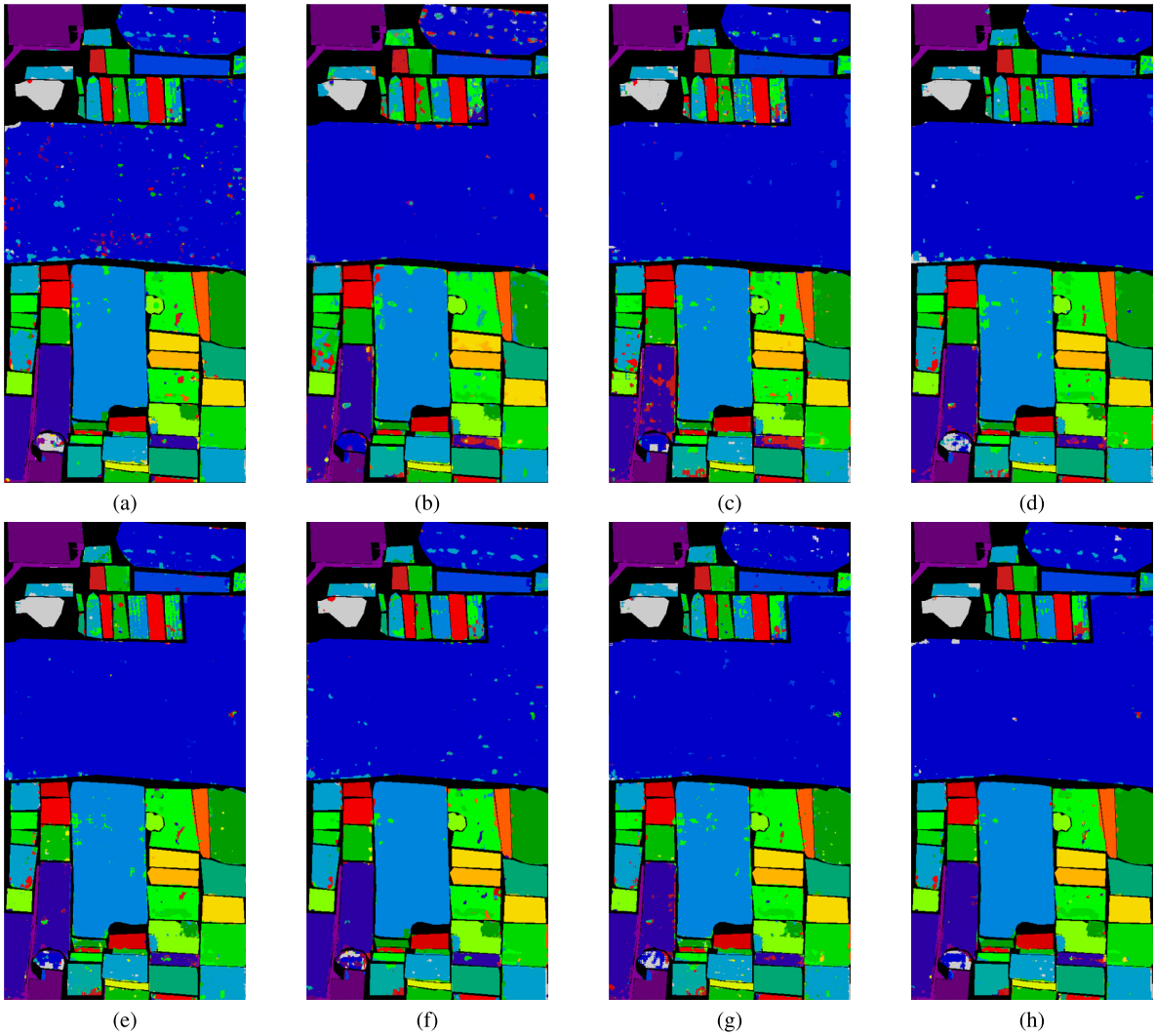


Fig. 6. Visual classification results with other methods on the HH dataset. (a) SSGRN. (b) SSRN. (c) SSFTT. (d) SQSFormer. (e) MambaHSI. (f) ConvMamba. (g) SpectralDiff. (h) Ours.

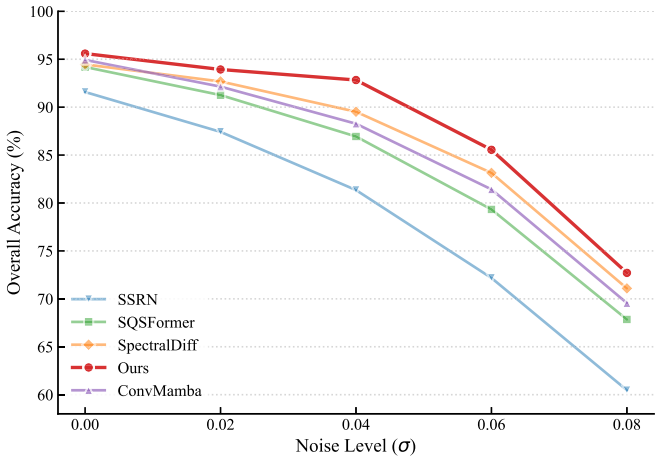


Fig. 7. OA curves under varying noise levels (σ) on the HH dataset.

B. Spatially Disjoint Sample Experiments and Result Analysis

To further evaluate the robustness of our HCD^2 model in practical remote sensing scenarios, a more challenging spatially disjoint experimental protocol is introduced in this work. Under this protocol, the training and testing samples are strictly separated in geographic space, to explore the cross-scene feature generalization ability of the model facing varied shooting regions, changing illumination environments and spatially heterogeneous land-cover distributions.

For the IP and PU datasets, we follow the spatially disjoint training and testing partitions provided by the IEEE GRSS Data and Algorithm Standard Evaluation (DASE) benchmark [60]. In the DASE protocol, the HSI scene is divided into nonoverlapping spatial patches, and these patches are assigned to either the training set or the test set, ensuring spatial separation between the two subsets. For the HH dataset, we construct spatially disjoint training and testing sets at the patch level following the same protocol. The original HSI image is divided into nonoverlapping patches with an identical spatial size. The generated patches, rather than individual pixels, are treated as the basic sampling units. We randomly select 5% of these patches for training, while the remaining 95% are exclusively assigned to testing. Consequently, no spatial patch is shared between the training and testing sets, avoiding the direct mixing of pixels from the same local patch across

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TABLE IV

QUANTITATIVE CLASSIFICATION COMPARISON ON THE PU DATASET (30 TRAINING SAMPLES PER CLASS). RESULTS ARE REPORTED AS THE MEAN AND STANDARD DEVIATION OVER TEN INDEPENDENT RUNS

Class No.	SSGRN[60]	SSRN[59]	SSFTT[42]	SQSFormer[43]	MambaHSI[15]	ConvMamba[47]	SpectralDiff[51]	Ours
1	82.40±5.87	75.72±12.31	92.40±5.05	93.91±0.94	94.35±1.73	97.19±1.41	86.15±8.42	**97.89±1.05**
2	96.03±1.84	83.26±26.03	97.77±1.96	96.60±3.28	96.43±1.92	97.94±1.53	98.39±1.12	**99.51±0.42**
3	93.72±2.56	92.90±4.54	94.56±2.53	89.66±6.70	93.39±5.35	94.04±2.75	96.57±1.84	**98.36±0.91**
4	**98.48±1.03**	87.50±6.45	94.93±2.32	98.02±1.47	89.75±3.60	96.93±1.27	76.40±8.17	96.01±2.08
5	99.47±0.36	94.84±5.15	99.88±0.08	99.97±0.04	99.99±0.02	99.95±0.10	99.54±0.38	**100.00±0.00**
6	**100.00±0.00**	98.33±3.28	99.10±1.06	98.81±0.28	98.60±0.93	99.46±0.91	98.30±1.43	99.12±0.69
7	99.92±0.08	97.57±1.83	**99.95±0.06**	99.85±1.11	96.50±3.13	99.69±0.25	99.08±0.52	99.67±0.27
8	87.43±5.42	71.83±27.95	93.75±3.11	85.49±10.34	94.40±2.93	90.94±4.35	**98.99±0.72**	91.88±4.26
9	92.47±3.76	75.75±4.90	98.91±0.48	99.67±0.21	99.36±0.62	99.65±0.26	91.17±5.74	**100.00±0.00**
OA(%)	93.85±1.21	84.29±13.70	96.54±0.86	95.52±1.55	95.74±0.90	97.29±0.89	94.78±1.73	**98.28±0.58**
AA(%)	94.43±1.08	86.41±6.24	96.80±0.65	95.78±0.79	95.86±1.11	97.31±0.60	93.84±1.36	**98.05±0.44**
$\kappa \times 100$	91.92±1.39	80.61±15.47	95.43±1.13	94.11±2.00	95.00±2.24	96.42±1.04	93.06±1.94	**97.72±0.66**

TABLE V

QUANTITATIVE CLASSIFICATION COMPARISON ON THE HH DATASET (30 TRAINING SAMPLES PER CLASS). RESULTS ARE REPORTED AS THE MEAN AND STANDARD DEVIATION OVER TEN INDEPENDENT RUNS

Class No.	SSGRN[60]	SSRN[59]	SSFTT[42]	SQSFormer[43]	MambaHSI[15]	ConvMamba[47]	SpectralDiff[51]	Ours
1	**97.64±1.08**	94.57±2.21	92.20±4.90	94.77±1.12	96.47±1.50	94.58±2.45	96.87±1.63	97.36±1.14
2	95.32±1.37	82.41±8.05	93.69±3.02	96.40±0.81	95.43±2.33	94.53±3.74	95.41±1.82	**97.18±1.09**
3	**94.77±1.91**	91.46±2.74	88.72±2.61	90.62±3.82	94.66±1.05	92.04±1.57	90.27±3.12	93.04±1.84
4	92.92±2.03	96.18±1.02	96.30±1.87	97.41±0.81	96.96±1.79	**98.09±0.68**	97.54±1.24	97.97±0.91
5	**99.79±0.23**	96.82±1.22	93.83±1.42	95.59±2.29	96.20±1.77	96.75±1.44	93.62±2.31	96.73±1.37
6	96.80±1.41	94.54±2.75	93.64±2.58	93.96±1.85	93.90±2.47	96.73±1.90	95.35±2.06	**97.81±1.08**
7	88.14±3.92	77.54±7.63	87.12±5.27	85.14±4.01	86.60±3.30	88.62±2.66	87.47±3.84	**90.63±2.41**
8	**98.27±0.86**	96.84±3.75	89.55±4.27	91.29±2.36	96.08±2.90	97.12±1.97	82.57±5.26	91.23±2.73
9	97.91±1.02	96.56±1.66	95.37±1.74	94.72±2.72	94.78±1.69	96.26±1.96	98.33±0.91	**98.92±0.54**
10	**96.54±1.46**	71.19±23.64	88.54±3.71	92.06±2.91	90.45±2.72	89.76±2.24	90.97±3.87	95.55±1.72
11	93.08±2.14	78.08±11.97	85.38±8.92	88.50±3.32	91.96±2.87	90.45±3.56	89.42±4.08	**93.42±2.19**
12	91.74±2.83	86.34±5.93	82.61±4.31	88.67±2.55	87.40±5.22	86.89±3.19	87.74±3.67	**93.10±2.32**
13	**80.77±4.31**	74.88±10.95	76.48±4.69	84.40±1.72	89.12±3.12	83.07±1.44	83.82±4.16	77.19±3.54
14	97.83±0.96	95.43±2.82	94.51±2.20	96.67±1.23	94.01±1.65	93.05±3.15	96.86±1.46	**98.91±0.58**
15	99.56±0.31	98.79±1.32	99.55±0.85	98.48±1.18	98.84±1.17	**99.84±0.17**	99.22±0.62	99.67±0.29
16	**99.85±0.14**	94.78±1.18	96.69±0.94	96.54±1.52	96.83±1.60	95.69±1.19	99.28±0.72	99.76±0.20
17	98.86±0.72	**99.97±0.05**	96.61±3.38	99.31±0.88	98.60±1.49	97.98±1.19	95.50±2.94	99.37±0.47
18	98.83±0.65	98.97±1.05	98.15±1.42	**99.50±0.40**	96.72±1.82	99.30±0.47	96.24±2.18	99.34±0.52
19	94.06±2.18	93.58±2.79	94.10±2.64	**96.11±2.09**	91.30±3.07	93.69±1.81	93.81±2.47	93.16±2.05
20	**99.49±0.42**	93.48±4.56	93.65±4.35	97.77±1.10	99.18±0.42	98.39±0.96	98.09±1.31	98.31±0.88
21	**100.00±0.00**	99.89±0.22	97.69±2.49	99.72±0.31	99.91±0.18	99.77±0.46	99.67±0.24	**100.00±0.00**
22	98.71±1.04	97.14±4.69	96.43±5.27	97.25±5.14	**99.54±0.51**	99.02±1.27	97.00±2.36	99.39±0.43
OA(%)	93.56±1.05	91.58±1.32	92.59±0.99	94.19±0.42	94.58±1.01	94.93±0.48	94.46±1.34	**95.59±0.51**
AA(%)	95.95±0.82	91.34±2.04	92.31±0.69	94.31±0.56	94.77±0.61	94.62±0.49	93.87±1.21	**95.82±0.46**
$\kappa \times 100$	91.99±1.17	89.46±1.65	90.71±1.21	92.70±0.52	94.00±1.88	93.62±0.61	93.04±1.58	**94.45±0.55**

TABLE VI

QUANTITATIVE CLASSIFICATION RESULTS UNDER THE SPATIALLY DISJOINT SAMPLE PARTITIONING PROTOCOL. RESULTS ARE REPORTED AS THE MEAN AND STANDARD DEVIATION OVER TEN INDEPENDENT RUNS

Method	IP			PU			HH		
	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
SSGRN[60]	78.42±1.87	70.15±2.41	76.54±2.03	82.35±1.63	83.41±1.82	77.63±1.74	79.14±2.16	81.55±2.03	74.32±2.24
SSRN[59]	75.10±2.34	72.34±2.81	72.88±2.56	79.56±2.07	77.12±2.46	74.05±2.29	76.22±2.73	73.41±2.95	71.50±2.84
SSFTT[42]	83.44±1.76	73.80±2.36	81.19±1.91	83.17±1.58	89.98±1.43	78.45±1.69	80.45±2.18	78.37±2.27	75.89±2.36
SQSFormer[43]	86.51±1.43	84.46±1.65	84.12±1.52	88.42±1.36	90.15±1.25	84.33±1.42	83.67±1.84	84.12±1.71	79.45±1.93
MambaHSI[15]	89.74±1.19	87.52±1.34	88.10±1.27	90.12±1.09	88.34±1.41	87.90±1.18	85.04±1.72	83.95±1.83	81.11±1.78
ConvMamba[47]	91.06±1.03	89.67±1.12	90.05±1.08	92.45±0.91	91.18±1.07	90.56±0.97	87.94±1.46	88.80±1.34	84.93±1.55
SpectralDiff[51]	87.23±1.68	85.55±1.82	85.67±1.76	93.10±1.24	93.85±1.15	92.41±1.29	88.15±1.63	86.22±1.77	85.54±1.69
Ours	**92.50±0.82**	**92.23±0.88**	**91.43±0.86**	**96.20±0.69**	**94.77±0.77**	**94.92±0.73**	**91.60±1.18**	**89.44±1.29**	**88.84±1.25**

the two subsets. This spatially disjoint partitioning protocol substantially reduces spatial dependence between training and testing samples and mitigates potential performance inflation caused by spatial autocorrelation and local spatial information leakage. The relevant outcomes are presented in Table VI.

From the results, we can find that our HCD^2 obtains optimal results on all three datasets, demonstrating the significant advantage of the generative classification framework in handling spatial shifts. Compared with SpectralDiff, which is also diffusion-based, HCD^2 consistently improves all evaluation metrics by approximately 3%~5%. The above results

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TABLE VII

ABLATION STUDY OF THE HCD^2 FRAMEWORK IN TERMS OF OA (%). RESULTS ARE REPORTED AS THE MEAN AND STANDARD DEVIATION OVER TEN INDEPENDENT RUNS

Model Variant	ASD	SSJTE	CDCD	Parameters (M)	IP	PU	HH
Baseline	×	×	×	0.08	76.09±1.84	89.40±1.37	79.80±1.92
w/o ASD	×	√	√	3.70	93.44±0.91	97.00±0.63	93.61±1.04
w/o SSJTE	√	×	√	3.64	91.99±1.06	96.63±0.71	93.13±1.12
HCD^2-Disc	√	√	×	0.19	91.19±1.18	94.81±0.86	88.43±1.31
Param-Matched Disc	√	√	×	3.74	93.02±0.73	96.31±0.52	92.41±0.88
HCD^2	√	√	√	3.73	**95.65±0.52**	**98.28±0.41**	**95.59±0.67**

show that using only the diffusion process is insufficient. The proposed SSJTE explicitly learns long-range topological relationships among land-cover categories, thereby providing more discriminative spatial priors for the diffusion decoder. This mechanism allows the model to gain better robustness and generalization under intense spatial heterogeneity, and finally delivers the optimal classification results across the three datasets.

The experimental results also show that our HCD^2 framework not only performs remarkably well under conventional random pixel sampling settings, but also exhibits superior robustness and accuracy under spatially disjoint scenarios that are more consistent with practical application conditions. As noted in Section IV-A, the spatially disjoint results provide a more conservative assessment that is free from the spatial autocorrelation inherent in the random-sampling protocol.

C. Ablation Study

Six model variants are designed for controlled comparison, as listed in Table VII. The baseline replaces the ASD with an identity mapping, the SSJTE with a 1×1 convolution projection, and the CDCD with a two-stage convolutional classification head performing single-step prediction. The w/o ASD and w/o SSJTE variants substitute one encoder component at a time while retaining the diffusion decoder. The HCD^2-Disk and Param-Matched Disk variants preserve the complete ASD+SSJTE encoder and attach a lightweight and a capacity-matched convolutional head, respectively (0.19M versus 3.74M), the latter matching the parameter budget of the full model so that the contribution of the diffusion decoder is assessed under capacity control.

As shown in Table VII, the full-version HCD^2 achieves the best performance on all three datasets, and each ablated module causes a consistent degradation (e.g., on IP, excluding the ASD, the SSJTE, or the diffusion decoder costs 2.21, 3.66, and 4.46 percentage points of OA, respectively). The Param-Matched Disk control further isolates the effect of the diffusion mechanism from capacity: the widened head improves over the lightweight HCD^2-Disk (e.g., + 3.98% OA on HH) yet still falls 2.63%–3.18% below the full HCD^2 with gaps far exceeding the standard deviations, confirming that the advantage of HCD^2 stems primarily from the diffusion-based decoding mechanism rather than from the increased decoder capacity.

Fig. 8 visualizes the error distributions of different model variants on the PU dataset. The baseline model produces massive misclassified pixels gathered along complex ground object boundaries. The model without the ASD module presents obvious classification errors in key local regions, which proves that the ASD module can effectively relieve local spectral aliasing problems. Meanwhile, the model removing SSJTE brings obvious block-like error distributions, indicating that long-range spatial topological modeling realized by SSJTE is essential to keep spatial consistency of classification results. Compared with the scattered errors of HCD^2-Disk, the Param-Matched Disk variant alleviates part of them thanks to its enlarged capacity, yet residual misclassifications still concentrate in spectrally ambiguous local regions. The complete HCD^2 applies iterative conditional denoising to further remove such pixel-level errors, balancing spectral discrimination with spatial continuity.

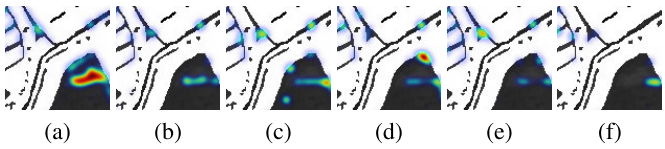


Fig. 8. Visualization of error distributions across different model variants. The gray-scale HSI base maps are overlaid with heatmaps, where colored regions represent misclassified pixels and transparent regions indicate correct predictions. (a) Baseline. (b) w/o ASD. (c) w/o SSJTE. (d) HCD^2-Disk. (e) Param-Matched Disk. (f) HCD^2.

D. Complexity Analysis and Parameter Analysis

1) Complexity Analysis: Efficiency quantifications on the PU scene are reported in Table VIII under the default sampling configuration of $T = 3$. The results provide a comprehensive comparison of the computational cost and inference efficiency of HCD^2 with the competing methods. With 3.73M parameters and relatively higher FLOPs than several discriminative baselines, HCD^2 incurs a moderate computational overhead due to the integration of the generative and adaptive spectral-aliasing blocks. Its inference time is higher than that of compact discriminative models such as ConvMamba, indicating that HCD^2 does not achieve the lowest computational cost among the compared methods. Nevertheless, compared with the generative SpectralDiff baseline, HCD^2 substantially reduces the inference time while maintaining competitive classification performance. This advantage is particularly relevant for generative HSI classification, where iterative sampling can introduce considerable inference overhead. Furthermore, the SSJTE module provides long-range topological priors that enable HCD^2 to maintain accurate category maps with a small number of sampling steps. Overall, the results demonstrate

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TABLE VIII

COMPUTATIONAL COMPLEXITY ANALYSIS ON PU DATASET (SAMPLING STEP = 3). OA IS REPORTED AS THE MEAN AND STANDARD DEVIATION OVER TEN INDEPENDENT RUNS

Metrics	SSGRN[60]	SSRN[59]	SSFTT[42]	SQSFormer[43]	MambaHSI[15]	ConvMamba[47]	SpectralDiff[51]	Ours
Parameters (M)	0.45	0.29	0.92	1.05	0.412	**0.18**	2.18	3.73
FLOPs(G)	0.32	0.25	**0.08**	0.23	39.18	0.07	4.25	6.90
Inference Time(s)	4.992	6.51	6.44	12.02	**0.206**	10.71	256.23	30.85
OA(%)	93.85±1.21	84.29±13.70	96.54±0.86	95.52±1.55	95.74±0.90	97.29±0.89	94.78±1.73	**98.28±0.58**

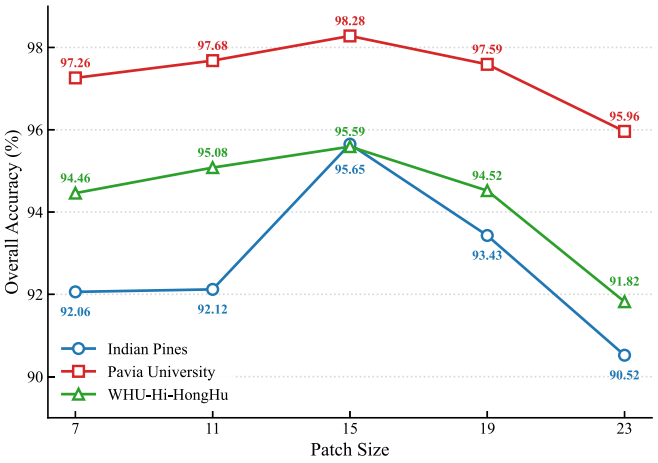


Fig. 9. Sensitivity analysis of the spatial patch size on the classification performance (OA) across three datasets.

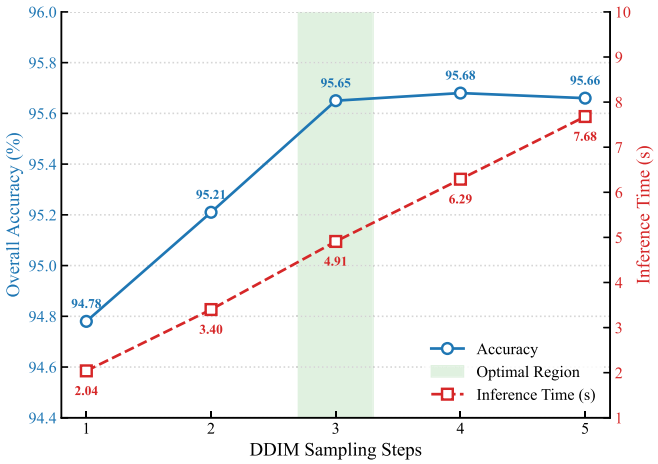


Fig. 10. Tradeoff between OA (%) and inference time (s) across different sampling steps T on the IP dataset.

that HCD^2 achieves a favorable balance between classification performance and computational cost, rather than providing a universally superior computational efficiency over all competing methods.

2) Patch Size: The patch size explicitly governs the network's contextual parsing capacity. As illustrated in Fig. 9, evaluating the parameter sensitivity of HCD^2 reveals a characteristic inverted U-shaped trajectory for OA across all evaluated benchmarks. Scaling the patch size from 7 × 7 to 15 × 15 secures progressive performance updates, where the IP, PU, and HH benchmarks culminate in peak accuracies of 95.65%, 98.28%, and 95.59%, respectively. Classification accuracy rises steadily with the enlargement of input patch size, which demonstrates that sufficient spatial topological prior information helps the generative model produce more coherent semantic classification results. Nevertheless, when the patch size exceeds 15 × 15, classification performance begins to decline gradually. Such performance drop is mainly attributed to redundant background interference and cross-category feature noise introduced by an oversized receptive field. Excessively large input patches will also lead to over-smoothed classification results, especially blurring fine dividing lines between adjacent ground object categories. Therefore, the patch size of 15 × 15 is selected as the optimal input setting for all subsequent experiments.

3) Sampling Step: The number of diffusion sampling steps T is a key hyperparameter that balances generation quality and computational cost. To explore its impact on classification performance, we carry out controlled experiments on the IP dataset. With the optimal 15 × 15 patch size fixed, we test model performance under different DDIM sampling steps ranging from 1 to 5, and the corresponding results are displayed in Fig. 10.

It can be observed that the classification performance maintains a steady and high accuracy when the sampling step T is set to 3. Further increasing sampling steps cannot bring apparent accuracy improvement, and even causes slight performance loss. This saturation trend is consistent with findings presented in previous diffusion model research for vision tasks [61], [62]. The slight performance decline with excessive sampling steps can be explained by sampling drift: there are inherent distribution differences between the data manifold used for model training and real samples used in inference, and more denoising iterations will enlarge such distribution gaps. During optimization, the network estimates the reverse trajectory conditioned on ground-truth-perturbed distributions; during inference, however, it must recursively filter noise from its own imperfect, parameterized prior predictions. This variance accumulation forces the generation trajectory to systematically deviate from the target manifold. Due to the cascading propagation of errors, this sampling drift becomes intensely acute at low-index time steps t, and is severely exacerbated when the initialization offset from the authentic distribution boundaries is substantial [20], [63].

We further examine the convergence behavior of the reverse process. For each sampling configuration, we record the intermediate prediction at every transition and measure its KL divergence from the ground-truth label distribution, averaged over all test pixels and ten independent sampling runs; the per-transition flip rate, i.e., the fraction of pixels whose argmax

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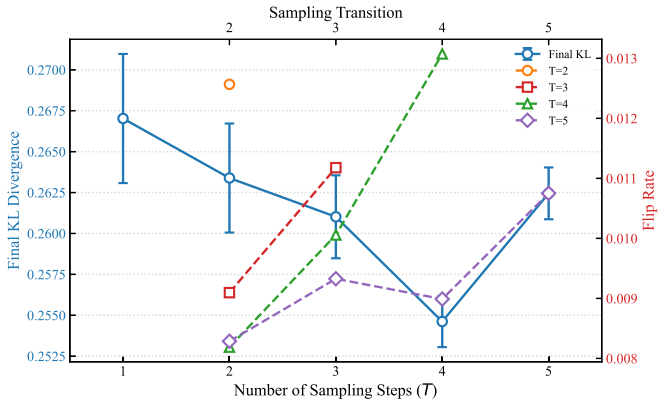


Fig. 11. Convergence analysis of the reverse diffusion process on the IP dataset: KL divergence between the intermediate prediction and the ground-truth label distribution along the reverse transitions (left axis) and the per-transition prediction flip rate (right axis) for different numbers of sampling steps T. KL values are averaged over all test pixels and ten independent sampling runs.

prediction changes, is tracked as well. As shown in Fig. 11, the final KL divergence decreases monotonically as T grows from 1 to 4 and rebounds at $T = 5$, closely following the OA trend in Fig. 10, whereas the flip rate remains around 1% throughout the trajectory and becomes more pronounced at the final transitions of larger T. The reverse process therefore never fully converges: beyond the saturation point, additional denoising steps no longer reduce the distributional discrepancy from the ground truth but mainly intensify the oscillations on ambiguous pixels.

While expanded sampling steps marginally refine the structural feature spaces, the accompanying error compounding and sampling drift bound continuous accuracy optimization. Given that computational latencies scale linearly with T, $T = 3$ is selected as the default configuration, striking an optimal Pareto compromise between classification precision and execution throughput across all scenario matrices.

V. Conclusion

This article introduces HCD^2, a generative framework that reformulates HSI dense classification as a conditional label-map denoising process. Building upon the label-space diffusion paradigm [20], the proposed model incorporates a dedicated spectral–spatial encoder and achieves progressive semantic inference through iterative conditional denoising. Our core architecture, comprising the ASD module for spectral decoupling and the SSJTE for topological manifold modeling, provides a high-fidelity spatial–spectral prior for the CDCD decoder.

All quantitative and qualitative results confirm the competitive advantages of the HCD^2 network against the discriminative and generative models in few-shot and spatially disjoint scenarios. The proposed method achieves stronger antinoise robustness in various test scenarios. We note that, consistent with the classical analysis of Ng and Jordan [19], discriminative classifiers may hold an advantage when training data is plentiful; the strength of the proposed generative formulation lies primarily in sample-scarce and cross-domain settings. Apart from solving the limitation of fixed semantic boundary division existing in conventional classification models, this research also offers feasible references and empirical bases for the follow-up diffusion-based applications in high-dimensional HSI analysis.

Code Availability

The source code will be made publicly available upon acceptance of this article.

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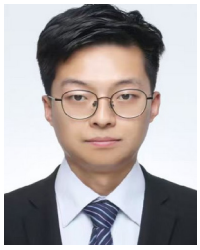
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